Aurelius
WEALTH MANAGEMENT

How we got here

A small team, a vibe coding mentality and an investment portal that makes money less daunting

The NTT Team, with Claude Code

THE SHORT VERSION

Vibe coding gave us speed. Principles made it shippable

The NTT Team set out to show that a vibe coding mentality, teamed with a small group of OutSystems developers, could rapidly craft a rounded investment portal.

The front end steered the design and architecture of the theme with Claude Code. A skilled OutSystems team took it and built the app in ODC.

PULL REQUESTS

186

merged into the theme

STYLE GUIDE

38

pages of documentation

CSS MODULES

36

one job per file

OSUI VARIABLES

63

bridged to our tokens

THE IDEA

Take what was out there, and improve it

THE QUESTION

Could a small team, with a vibe coding mentality, build something better than what is out there?

Why finance

FinTech experience in the team and NTT's banking division made investment portals familiar ground.

Why vibe coding

To craft a product rapidly, without giving up the quality a wealth client expects.

Why OutSystems

To show the approach could reach a real application in the hands of skilled developers, not stop at a demo.

THE AMBITION

Nobody likes talking about money

> *"it needs to feel ultra modern and simple, not intimidating"*
>
> Jamie Scott, front end

Engage, don't intimidate

A rounded investment portal that draws people in, rather than scaring them away from the subject.

Room to breathe

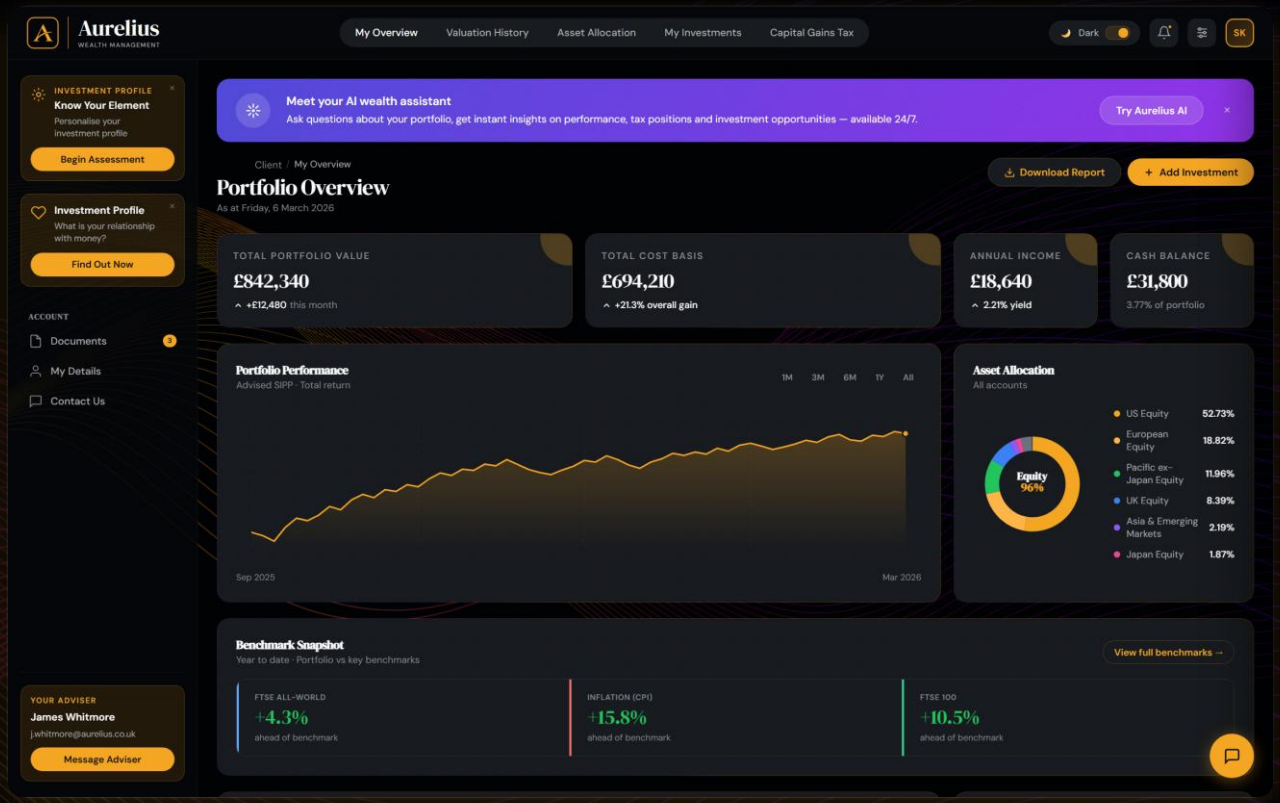
Clear the clutter and let the content breathe, so nobody feels overloaded.

Considerate by design

Real care around sensitive questions, such as family or bereavement.

Platinum for everyone

A premium experience that never feels like a tool for the elite.



The visual language: calm, dark and spacious

THE RESEARCH

We studied what was out there, then asked for better

What the NTT Team brought

The bar to beat

Bootstrap, Tailwind and Material as examples of great style guides. “we need better than that”.

Design direction

Award-winning sites and studios, and leading consumer investing experiences.

Platform knowledge

What OutSystems provides, so the prototype and the application would match.

What Claude Code brought

Gap analysis

A ranked list of what an investment portal should offer and what was still missing.

Root causes

Why the theme’s CSS kept losing on OutSystems: layers, not specificity.

Concepts to react to

Ideas pitched first, then challenged and refined before anything was built.

THE TEAM

A small team, with clear roles

The NTT Team A shared idea and a shared standard

Front-end direction

Steers the design and the architecture of the theme, and holds the work to a standard.

GIVES **Direction and standards**

→

Claude Code

Builds the theme, the HTML prototype and the documentation at speed, under that direction.

GIVES **Theme and style guide**

→

OutSystems developers

A skilled team takes the theme and builds the application in ODC.

GIVES **The application in ODC**

Vibe coding did not replace skilled developers. It was teamed with them.

THE WAY OF WORKING

Human direction, AI build, constant challenge

Direct

Set the brief, the standard and the constraints first.

“remember how we work, clean CSS over JS, repeatable reusable tokens and components”

Build

Claude Code builds on a branch and opens a pull request.

“throw the next PR my way please”

Challenge

Review it in the browser and send it back until it is right.

“no that’s horrible, roll that back, lets add a bottom gold border and maybe a subtle bottom glow”

Corrections became standing rules

Quotes: Jamie Scott, front end

Features need a branch and a PR

Written after a second direct push to main

Never edit bundle.css directly

It is generated, and a deploy wipes edits

Prefix classes with .site-

OutSystems UI owns the generic names

No utility classes

Name classes for what the thing is

PRINCIPLE · CSS

CSS over JS, because CSS is what travels to OutSystems

BEFORE

```
toggleTheme()
/* JS flips an attribute */
[data-theme="dark"] { … }
```

AFTER, NO JAVASCRIPT

```
<input type="checkbox" role="switch"
       class="theme-checkbox">

html:has(.theme-checkbox:checked) {
  color-scheme: light;
}
--bg: light-dark(var(--neutral-1), var(--neutral-10));
```

> *"any custom JS will be difficult to implement in OutSystems"*
>
> Jamie Scott, front end

What CSS carries

Theming
Light and dark from one set of tokens

State
:has() and native inputs, not scripts

Layout
Grid that holds up inside OutSystems wrappers

Motion
Native transitions that respect reduced motion

PRINCIPLE · HTML

HTML that means something before it looks like anything

```
<header role="banner">
  <nav class="topbar-nav"
       aria-label="Primary">
</header>

<aside class="site-sidebar"
       aria-label="Site navigation">
  <div role="group"
       aria-labelledby="nav-account">
</aside>

<main class="main">
```

Landmarks, not divs

A real banner, navigation and main area, so the structure is clear to every user.

A real heading structure

Semantic headings across the portal, treated as a piece of work in its own right.

Native elements first

Built-in form and output elements before custom widgets, for less code and better behaviour.

Names that survive change

Class names describe what a thing is, not where it sits.

PRINCIPLE · ACCESSIBILITY

Accessible from the start, not audited at the end

ONE TOKEN, 12 FAILURES

`--text-3 in dark mode`

3.55:1 → **5.61:1**

Fails WCAG AA
Wealth Management
#6B6F76

Passes WCAG AA
Wealth Management
#8C9099

Lowest ratio shown for each value. All 12 Lighthouse failures traced to this single token, so a single change cleared them.

Lighthouse runs after every deploy.

And animated borders stop for anyone who asks for less movement.

The standard, held in review

"we're accessible from the go remember"

Input borders were raised to meet 3:1, creating a dedicated token.

"we can't hide that away for accessibility"

A modal close button was made gold, so it could not be missed.

"for a screen reader this would be even weirder"

A calculated result was rethought, not just restyled.

"nested scrollbars aren't great for accessibility"

Layouts were reworked to avoid scrolling inside scrolling.

PRINCIPLE · DESIGN TOKENS

Three seed colours drive the entire palette

SEEDS

#F5A623
primary

#FFB547
secondary

#6B6F76
neutral

GENERATED SCALES

--brand-50 to 900



--neutral-0 to 10



oklch(from var(--seed-primary) l c h)

NAMED TOKENS

--gold
--bg
--bg-card
--text-1
--text-2
--text-3
--border-input

OUTSYSTEMS READS

--color-primary
--color-error
--color-neutral-0
--color-neutral-10
--font-size-base
--border-radius-soft

Change one seed and the whole system follows, in the prototype and in the OutSystems variables alike. A new theme becomes an edit, not a project.

One bridge file teaches OutSystems to speak Aurelius

OUTSYSTEMS UI ASKS FOR

- `--color-primary`
- `--color-error`
- `--font-size-base`
- `--font-family-heading`
- `--border-radius-soft`

→

`osui-bridge.css`

A one-way adapter. It only sets OutSystems variables from Aurelius tokens.

→

AURELIUS PROVIDES

- `--gold`
- `--red`
- `--fs-base`
- `--font-serif`
- `--radius-item`

VARIABLES FIRST MAPPED

63

CATEGORIES

10

AURELIUS TOKENS MODIFIED

0

One stylesheet, two destinations



Review is the release gate

Features reach main through a reviewed pull request, and merging is what updates the theme.

The CSS must be self-contained

ODC never sees the prototype's page heads. Fonts and images have to load from the CSS.

The prototype is the test bed

Designs are proven on the prototype before the OutSystems team builds with them.

Building in ODC reshaped the theme

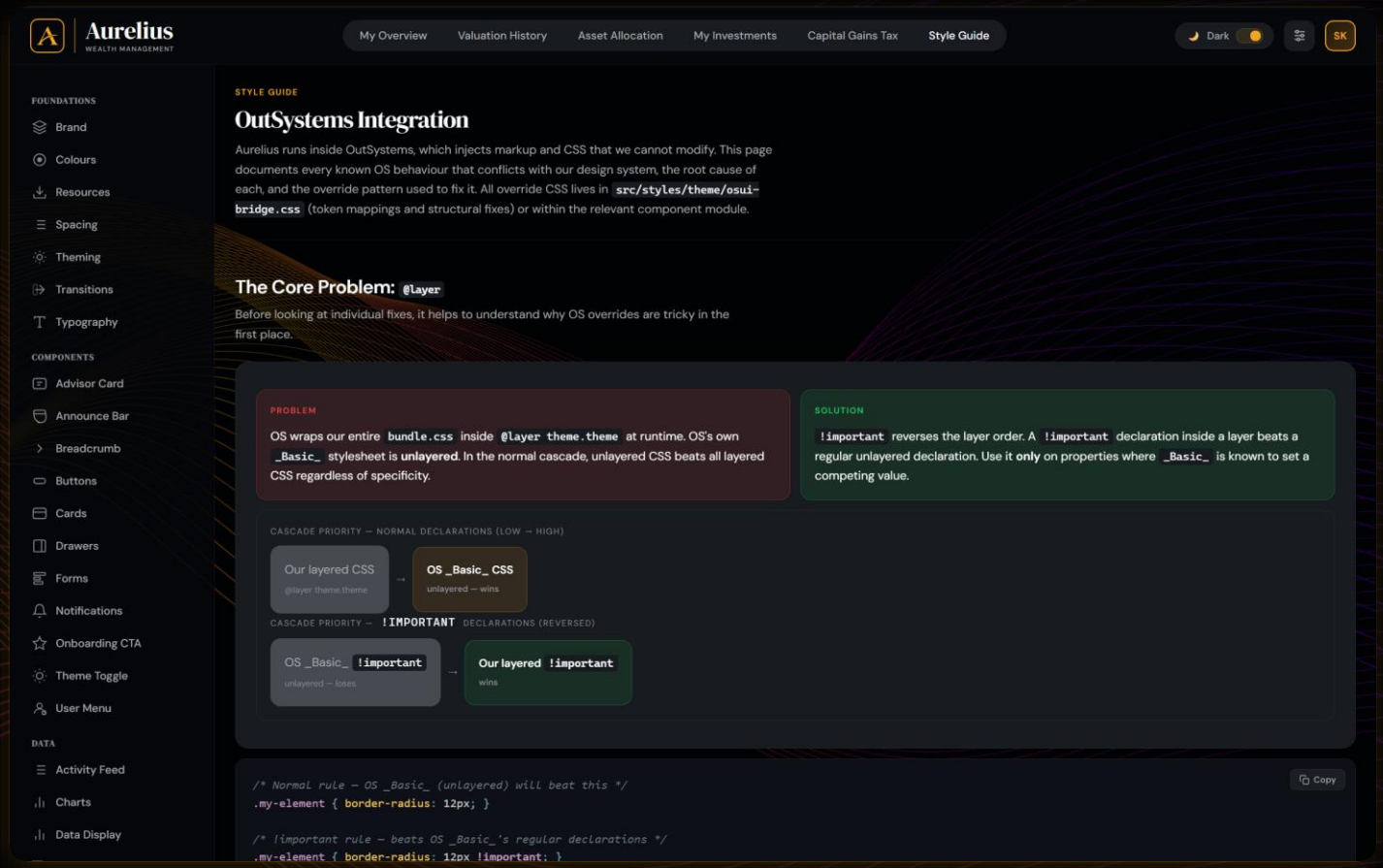
WHAT WE FOUND	WHY IT HURT	WHAT WE CHANGED
Blocks arrive wrapped in extra divs	A flexbox header stopped distributing its children	A three-column grid, with display: contents to flatten wrappers
OutSystems UI owns .sidebar	Our sidebar inherited an off-canvas position	Renamed to .site-sidebar, then a .site- prefix rule
Our CSS sits in a layer, _Basic_ does not	Unlayered rules always win, however specific ours are	Targeted !important, only where _Basic_ is shown to win
No Popover API support	Drawers and menus had no open state to style	A parallel .is-open class beside every :popover-open
Page heads never reach the platform	The wordmark font fell back to Georgia	@font-face rules moved into the bundle itself

The style guide became the shared language

the guide needs to be complete documentation so any developer can access it and be able to build a component or a page

Every page carries a live demo, copy-ready HTML, a token table, do and don't guidance, accessibility notes and OutSystems callouts where relevant.

38 pages of living documentation for developers and designers



A class name should say what a thing is

247

.u-* utility classes added by the AI's first attempt at removing inline styles

Rejected on review. All gone before merge.

"they leave you with situations where if things change the class name becomes detached from the use"

Jamie Scott, front end, on utility classes

```
<button class="ais-mode-btn
  u-py-xs u-px-base u-radius-card
  u-border-0 u-bg-none u-text-2
  u-fs-sm u-pointer">
```

```
<button class="ais-mode-btn">
/* the component owns its styling */
```

Named by role

Each class belongs to a component and lives in that component's partial.

Media queries in the component

Responsive and state changes stay in CSS, out of the markup.

Mapped to OSUI widgets

The guide says which widget each supporting class applies to.

WHERE IT STANDS

What came out of it

A portal

A rounded investment portal that wants to engage people, built in ODC by a skilled team of OutSystems developers.

A theme

A documented, token-driven theme with its own bridge to OutSystems UI, built for reuse beyond this one application.

A way of working

A vibe coding mentality teamed with front-end standards and platform skills, in a team small enough to move quickly.

The portal was the proving ground. The theme and way of working can be reused.

WHERE IT STANDS

What is still open

Newest concepts live in the prototype

The latest proof-of-concept work has not been carried across to OutSystems.

Concept

Accessibility is still owed on proof-of-concept work

A conscious decision for a demo. The full run-through and mobile pass still need doing.

Owed

The latest theme guidance needs proving in ODC

The cascade was reasoned through rather than run on the platform.

To verify

THE LESSONS

What we would carry forward

Set the standard before the speed

AI multiplies whatever direction it is given. Experienced judgement cannot be skipped.

Team AI with skilled people

Vibe coding gets you moving. Skilled developers get you to a real application.

Make every correction permanent

A correction that becomes a written rule never has to be made twice.

Prefer what the platform gives you

Native HTML and CSS meant less code to carry across to OutSystems.

Design for the destination

Know how the final host treats your code, then test against it early and often.

Document as you build

A living style guide is what lets a team share the work.

simple to use · platinum level experience · CSS over JS · cutting edge · above all accessible

“this is not just a platinum wealth management experience, it is a platinum grade theme”

Jamie Scott, front end

APPENDIX

What this account rests on

PART OF THE STORY	RESTS ON
The idea, the ambition and the team's roles	The NTT Team's own account
The theme, prototype and style guide	Repository history, pull requests and surviving session transcripts
The application built in ODC	The team's own account. It is not in the theme repository and was not inspected for this deck.

Not verified here

The running ODC application. No Lighthouse score history exists beyond PR #24. Research from the earliest months was not recorded.

How quotes were handled

Front-end quotes are Jamie Scott's, with typos and punctuation tidied and some shortened. PR write-ups were drafted by Claude Code.

Credits

The onboarding experience was built to a U Impact reference design. Screenshots use demo figures, with the client name masked.